MILITARY SPECIFICATION

TANK AND PUMP UNIT PMU-29/E, JET OIL SERVICING 3-GALLON CAPACITY, PORTABLE

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 3-gallon capacity, jet oil servicing tank and pump unit, designated PMU-29/E.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Box, Shipping Fiberboard.
PPP-T-60	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-L-7808	Lubrication Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-8188	Corrosion-Preventive Oil, Gas Turbine Engine, Aircraft Synthetic Base,
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant.
MIL-P-15024/6	Plate, Identification, Equipment.
MIL-L-23699	Lubrication Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-N-25677	Nozzle, Pressure Lubricating Oil Servicing, Locking.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipment).
MIL-STD-471	Maintainability Demonstration.
MIL-STD-808	Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Water-Proofing, with Appropriate Test Methods.
MS24475	Nozzle, Pressure Lubricating Oil Servicing, Locking.
MS24476	Adapter, Pressure Lubricating Oil Servicing, Aircraft.
MS28720	Filter, Aircraft Hydraulic Line Type, 3,000 PSI.

* 2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the data of the solicitation.

DRAWING

AIR FORCE

7545352	Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground Support Equipment.

AIR FORCE DRAWINGS

AIR FORCE-NAVY AERONAUTICAL

AN6235	Filter Element - Hydraulic Replaceable Micronic Line Type.

* (Copies of specifications, standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-3951	Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing (see 4.4 and 6.5).

3.2 Components. The tank and pump unit shall consist of the following major components:

Supply tank (see 3.7.1).

Hand pump (see 3.7.2).

Filter assembly (see 3.7.3).

Dispensing hose assembly (see 3.7.4).

Return hose assembly (see 3.7.5).

Overflow tank (see 3.7.6).

Three-wheel dolly (see 3.7.7).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

* 3.3.1 Standard parts. Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there are no suitable corresponding standard parts in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Materials. All materials shall be compatible with oil conforming to MIL-L-7808, MIL-C-8188, and MIL-L-23699. All materials shall be of uniform quality and condition and free of cracks, voids, or other defects which may adversely affect the appearance, strength, performance, durability or endurance of the finished tank and pump unit. Wood shall not be used.

3.4.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them, When used and not hermetically sealed, they shall be treated with a fungicidal agent, However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

* 3.4.2 Metals. Metals shall be either non-ferrous or stainless steel. Cadmium (including plating) and magnesium (including alloys) shall not be used.

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3.4.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

* 3.4.3 Elastomers. Except for dolly wheel tires, elastomers not otherwise specified shall be selected from those elastomers listed in 4.6.3 Elastomer compatibility test of MIL-L-7808. Dolly wheel tires shall be MIL-L-7808 oil resistant.

* 3.5 Design and construction. The tank and pump unit shall dispense oil to aircraft lubricating oil systems requiring pressure servicing with synthetic base lubricating oils. The unit shall permit convenient, one-man operation. The unit shall maintain oil initial cleanliness with respect to paarticulates and water under all operating environments and circumstances. Unit exterior shall have no feature that might trap fluids. Unit parts shall not work loose in service and shall withstand the stresses, jars, vibrations and other conditions incident to shipping, storage, movement and operation. Parts subject to loss shall be retained to the unit by means of a 1/32-inch diameter wire rope with clear nylon jacket. Components shall be appropriate for military flight line equipment.

3.5.1 Reliability. The tank and pump unit shall have a minimum MTBF of 27,000 pump cycles with a minimum reliability of .9991 for a mission time of 25 pump cycles at a confidence factor of .90 (see 6.3.1, 6.3.2, 6.3.3, and 6.4).

3.5.2 Maintainability. The design and construction of the tank and pump unit shall comply with the maintainability requirements specified in MIL-STD-470.

* 3.5.3 Component removal. The tank and pump unit shall be so designed that all major components (see 3.2) may be easily removed, replaced, and operated by personnel wearing arctic, gloves.

3.5.4 Handtools. The tank and pump unit shall be so designed and constructed that assembly, and disassembly can be performed with the use of common handtools.

* 3.6 Performance. At 70° +5°F ambient temperature and with its MS24475-1 delivery nozzle connected to a MS24476-1 adapter, the unit shall dispense oil conforming to MIL-L-7808 at a rate of not less than 1 gpm with not more than 50 strokes or revolutions per minute of the pump handle. At not more than -40°F and with its MS24475-1 delivery nozzle connected to a MS24476-1 adapter, the unit shall dispense MIL-L-7808 oil at a rate of not less than 0,25 gpm with not more than 50 strokes or revolutions of the pump handle and with not more than 50 pounds exerted at the pump handle grip. At not more than -40°F and with not more than 50 pounds exerted at the pump handle, the unit shall develop not less than 300 psig deadhead pressure of MIL-L-7808 oil. The unit shall operate under all conditions and environments without leaking.

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* 3.6.1 Environmental conditions. The tank and pump unit shall operate under and satisfactorily resist the following conditions and shall maintain oil initial cleanliness.

* a. Temperatures ranging from -40° to +125°F.

b. A relative humidity of 95 to 100 percent at an operating temperature of 125°F and a storage temperature of 160°F.

c. Atmosphere containing salt-fog.

d. Fungus as encountered in tropical climates.

e. Sand and dust particles as encountered in normal and desert operations.

f. Rainfall as encountered in any locale.

3.6.1.1 Storage temperatures. The tank and pump unit shall withstand storage temperatures ranging from -80° to +160°F.

* 3.6.2 Pressures. All components downstream of the pump assembly, including the fill nozzle, shall withstand a test pressure of 500 psi or that deadhead pressure produced with a force of 150 pounds at the pump handle, whichever is greater. The supply and overflow tanks shall withstand an internal pressure of 10 psi or 200 percent of the operating pressure, whichever is greater.

3.6.3 Mobility. The tank and pump unit, complete with all components and mounted on the 3-wheel dolly specified in 3.7.7, shall meet the following requirements:

a. Being carried by two men and placed in its upright position (see 3.7.7).

b. With a full supply tank, be towable at an average speed of 5 mph (maximum speed of 7.5 mph) with the capability of negotiating right- and left-circle turns at the maximum cramping angle of a towing vehicle with a 10-foot turning radius.

c. With a full supply tank, be towable at an average speed of 5 mph (maximum speed 7.5 mph) over .75 to 1-inch high, 60 degree angle obstructions without damage to the structure or components.

* d. With a full supply tank, withstand being dropped from a height of 1 foot onto a concrete floor from various tilted positions so that one wheel will impact before the other wheel without damage to the structure or components.

e. Remain in a natural upright position when a force of 50 pounds is applied within 15 degrees of vertical at the center of the pump handle grip.

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3.7 Details of components.

* 3.7.1 Supply tank. The supply tank shall have a minimum opening 6 inches in diameter in the top to provide for pump assembly mounting and easy access for interior tank cleaning. It shall have a minimum usable capacity of 3 gallons. All joints and seams shall be smooth and free from voids, sharp edges, and sharp bends. All bends and corner joints shall have at least a 0.25 inch radius which will prevent dirt and residue accumulation and facilitate cleaning of the tank.

* 3.7.1.1 Filler cap. A filler cap shall be provided and shall be designed to prevent leakage of air or oil. The filler cap shall be easily installed or removed by hand. Means shall be provided to prevent loss of the cap when opened. The minimum inside clearance diameter of the filler opening shall be 1.50 inches. If cap is threaded, then threads shall be female.

* 3.7.1.2 Air vent. An air vent shall be provided on the tank to prevent pressure or vacuum lock when oil is being pumped into or out of the tank. This provision shall not, however, preclude pressurization of the tank should such a design be required. The air vent shall prevent loss of oil should the assembly upset and shall preclude entrance of water under rain conditions. A 100-mesh (minimum) screen shall be incorporated into the vent.

3.7.1.3 Adapter. An MS24476-1 adapter shall be provided on the tank to allow the tank to be filled by another tank and pump unit which is equipped with mating nozzle.

* 3.7.1.4 Sight glass assembly. A liquid level sight glass assembly graduated in 0.5 pint increments a minimum of .1875 inch apart shall be provided on the supply tank. The graduations shall indicate the volume of oil pumped from tile tank rather than the amount of oil remaining in the tank. The sight glass shall be removable for cleaning. The lower visible end of the sight glass shall not be below that oil level needed for positive pumping of oil so that no air is pumped if oil is visible in the sight glass. A movable pointer shall be provided so that the level of oil can be marked. If the sight glass proper is fabricated from a non-vitreous material, then evidence shall be submitted in the first article test report that the sight glass material is compatible (no clouding, no crazing, etc) with MIL-L-7808 oil.

3.7.1.4.1 Guard. The sight glass shall be protected by a metal guard. The guard shall consist of a rigid metal tube with two openings spaced 180 degrees apart. The operator shall view the oil level in the sight glass by looking tangentially to the tank through the guard openings. The guard shall have sufficient strength to resist flexure when a 25 pound static force is applied at the midpoint of the long axis.

* 3.7.1.5 Mounting. The supply tank shall be securely mounted on the 3-wheel dolly specified in 3.7.7. The tank with pump assembly, filter, and dispensing hose shall be removable as an assembly without the use of handtools.

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3.7.1.6 Air relief valve. If the system is pressurized, an air relief valve shall be provided. The valve shall be set to relieve at pneumatic operating pressure and shall also serve as a means of depressurizing the supply tank after the servicing operation has been completed.

* 3.7,2 Hand pump. A hand-operated, self-priming pump shall be incorporated into the design of the supply tank, The pump shall meet the pumping requirements specified in 3.6. An adjustable leverage principle may be incorporated to meet the pumping requirements under the varied temperature conditions. The pump shaft shall be fitted with an exclusion seal preventing the entry of dust and water into the unit.

3.7.2.1 Pressure relief. A means of relieving pressure in the discharge hose shall be provided in the pump assembly so that the filter and discharge line remain flooded with oil between servicing, but pressure is relieved.

3.7.2.2 Pump handle. The pump handle shall be so located that it is conveniently accessible and the operator has sufficient operating room during pumping. The pump handle shall serve as a means of carrying the supply tank when it is removed from the 3-wheel dolly, and shall have a grip so designed that it can be easily grasped by personnel wearing heavy, arctic winter gloves.

* 3.7.3 Filter assembly. A filter assembly shall be installed between the outlet of the pump and the inlet to the discharge hose. The filter assembly shall consist of a case conforming to MS28720 and a filter element conforming to AN6235. The filter assembly shall not have a bypass. All O-rings and gaskets shall be compatible with oil conforming to MIL-L-7808.

* 3.7.4 Dispensing hose assembly. A 6-foot length of size 10 assembly hose conforming to MIL-H-8795 shall be provided and installed at the discharge end of the filter assembly. A nozzle in accordance with MS24475-1 shall be installed at the discharge end of the hose.

* 3.7.5 Return hose assembly. A 6-foot length of size 12 hose assembly conforming to MIL-H-8795 shall be provided. One end of the return hose shall be equipped with a sight glass fastened to the overflow tank. The free end of the return hose shall be provided with a nozzle in accordance with MS24475-2 for attachment to the engine overflow adapter of the aircraft. A plug, rigidly fastened to the unit near the adapter specified in 3.7.1.3, shall be provided to keep the nozzle clean when not in use and to provide for positive storage of the return hose end.

* 3.7.6 Overflow tank. The overflow tank shall have a minimum opening 6 inches in diameter in the top cover to provide easy access for interior tank cleaning. The tank shall be attached to the 3-wheel dolly frame with quick-release fasteners so that it can be removed without the use of handtools. The tank shall have a minimum capacity of 2 gallons. All joints and seams shall be smooth and free from voids, sharp edges, and sharp bends. All bends and corner joints shall have at least a 0.25 inch radius which will prevent dirt and residue accumulation and facilitate cleaning of the tank.

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* 3.7.6.1 Sight glass assembly. A liquid level sight glass assembly shall be provided on the overflow tank and graduated in 0.5 pint increments a minimum of .1875 inch apart. The graduations shall indicate the volume of oil in the tank. A movable pointer shall be provided so that the level of oil can be marked. The sight glass shall be removable for cleaning. The sight glass shall be provided with a metal guard as specified in 3.7.1.4.1.

3.7.6.2 Drain opening and cap. A minimum l-inch diameter drain opening with a cap designed to prevent leakage of oil shall be provided on the top of the tank. The drain opening shall be located near the side of the tank so that oil can be easily poured from the tank. Means shall be provided to prevent loss of the cap.

3.7.6.3 Air vent. A ball-type air vent shall be provided on the overflow tank to prevent pressure lock when oil is flowing into the tank. The vent shall prevent loss of oil should the assembly upset and shall preclude entrance of water under rain conditions. A 100-mesh screen shall be incorporated into the vent.

3.7.6.4 Handle. A handle shall be provided on the overflow tank so that when the tank is in a filled condition, it can be conveniently carried and easily emptied.

* 3.7.7 Three-wheel dolly. The tank and pump unit and overflow tank shall be mounted on a 3-wheel dolly consisting of a welded tubular aluminum frame with two non-swivel trailing wheels, a forward 360-degree swivel wheel, and a towbar. The dolly shall have a minimum ground clearance of 2 1/2 inches. Positive seating receptacles and hold-down mechanisms for the supply and overflow tanks shall be provided on the frame. If the frame does not inherently provide convenient gripping places for lifting of the complete unit with a full supply tank by two men, then lift handles shall be provided.

* 3.7.7.1 Wheels. Wheels shall be 6-inch (minimum) diameter with either solid or semi-pneumatic, tires. Wheels shall have anti-friction bearings.

* 3.7.7.2 Towbar. The towbar shall provide for convenient towing and maneuvering of the dolly by a person in a normal walking position. The towbar shall be fitted with a cross bar handle.

3.7.7.3 Hose storage. Positive-holding provisions shall be incorporated on the tanks for storage of both the fill and overflow hoses. The fill hose nozzle shall be connected to the adapter specified in 3.7.1.3 and the overflow nozzle shall be connected to the plug specified in 3.7.5 for securing of the hoses during storage.

* 3.7.7.4 Stabilization. Means shall be provided for locking both the front swivel and wheel so as to stabilize the unit during oil pump operation. The lock shall be easily and effectively engaged and disengaged by either a single foot operated lever or by moving the towbar to and from its stowage position, the towbar shall not interfere with pump handle operation.

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3.7.8 Fill nozzle adapter. An MS24476-1 nipple with a 3-inch long pipe extension shall be provided for connection to the fill nozzle (see 3.7.4) for servicing of aircraft with gravity-fill oil tanks. Dust caps shall be provided for both ends of the adapter.

* 3.7.8.1 Storage provisions. A dustproof gasketed metal box shall be provided for positive storage of the nozzle adapter supplied with the unit.

* 3.8 Dimensions. The storage dimensions of the tank and pump unit, excluding the towbar shall not exceed 30 by 32 by 35 inches high. The operational envelope of the tank and pump unit shall not exceed 30 by 32 by 37 inches high.

3.9 Weight. The dry weight of the tank and pump unit shall not exceed 96 pounds.

* 3.10 Finish. The unit shall be finished in accordance with MIL-STD-808 for type I exposure as supplemented by Air Force drawing 7545352 except that interior oil wetted surfaces shall be neither primed nor painted and except that interior oil wetted surfaces of components fabricated from aluminum alloy shall be anodized per MIL-A-8625 either type I or II according to the particular alloy except that components fabricated from either alloy 6061 or 6063 need not have their interior oil wetted surfaces either anodized or any other finish.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Operational markings.

* 3.12.1 Identification. The unit shall be provided with an identification plate conforming to MIL-P-15024/6.

* 3.12.2 Operating instructions. The unit shall be provided with an operating instructions plate that matches the identification plate with regard to style, color and material.

* 3.13 Workmanship. The tank and pump unit, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, welding, brazing, painting, riveting, tightness of screw assemblies, and freedom of parts from burrs and sharp edges. Workmanship shall be free of defects that might adversely impact appearance, performance and durability.

3.13.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the Interchangeability, operation, or performance of the tank and pump unit, they shall be held or limited accordingly,

3.13.2 Riveting. Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

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3.13.3 Cleaning. The tank and pump unit, especially the tanks, shall be thoroughly cleaned of loose, spattered, or excess solder; metal chips, weld spattering; and other foreign material during and after final assembly. Burrs and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the tank and pump unit shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance tests (see 4.5).

4.3 Test conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70°F). The MS24475-1 delivery nozzle shall be connected to a MS24476-1 adapter during all flow test.

4.3.1 Test fluid. The test fluid shall be lubricating oil conforming to MIL-L-7808.

* 4.4 First article testing.

4.4.1 Test sample. One tank and pump unit shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831, except an authorized Government quality control representative shall countersign all data sheets and the report. The following shall be Incorporated as a part of the test report:

a. Eight-by-ten-inch black and white photographs of both sides of the tank and pump unit.

b. 35-mm colored slides of three-quarter, right-front views and three-quarter, left-rear views of the complete tank and pump unit.

c. A list, by component manufacturers, stating the metals used in fabrication of components.

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d. A list of all elastomeric components not otherwise specified showing that they are comprised of materials listed in 4.6.3 of MIL-L-7808. List shall include wheel tire material with a statement concerning MIL-L-7808 oil resistance.

e. A list of all cleaning and finishing procedures of MIL-STD-808 and AF drawing 7545352.

f. Completed checklist required by 4.6.1.

4.4.2.1 Reliability and maintainability information. The following information shall be incorporated as an attachment accompanying the first article test report (same quantity and quality of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating times, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made, and the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.3 First article tests. The first article tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests (see 4.5.1).

b. Sampling plan and tests (see 4.5.2).

4.5.1 Individual tests. Each tank and pump unit shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. pressure test (see 4.6.2).

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4.5.2 Sampling plan and tests.

4.5.2.1 Periodic sampling. One tank and pump unit shall be selected at random from each group of 20 or fraction thereof produced and subjected to the operation test specified in 4.6.3.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

* 4.6.1 Examination of product. The first article unit shall be examined to confirm adherence to such specified requirements as configuration, design details, use of specified parts and marking. This examination shall consist of visual inspection of the unit, simple measurement of unit dimensions, weighing of the unit, examination of certificates of compliance on purchased materials and components, examination of the lists required by 4.4.2.c, d and e and review of unit manufacturing drawings as required. This examination of the unit shall be accomplished using a contractor developed check list that shall identify all specific unit attributes for which examination is made, how conformance is determined and results obtained. This contractor developed check list shall additionally be used to accomplish acceptance test examination of product.

* 4.6.2 Pressure test. The supply tank (including fittings) shall be subjected to an internal air pressure of 10 psi or 200 percent of the operating pressure, whichever is greater, for 10 minutes and the overflow tank (including fittings) shall be subjected to an internal pressure of 10 psi for a duration of 10 minutes. A mercury manometer shall be connected to the tank, or the tank shall be submerged in water to check for leakage. Any evidence of leakage shall be cause for rejection. The portion of the tank and pump unit on the discharge side of the pump, to and including the dispensing hose nozzle, shall be subjected to an internal hydrostatic pressure of 500 psi or that pressure produced by a force of 150 pounds applied at the pump handle, whichever is greater, for 5 minutes and 20 psi for 5 minutes. Any evidence of leakage shall be cause for rejection.

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* 4.6.3 Operation. The supply tank shall be filled with lubricating oil conforming to MIL-L-7808. With the oil and the unit at a temperature of 70° +5°F, the tank and pump unit shall discharge oil from the end of the fill nozzle at the rate of not less than 1 gpm with not more than 50 strokes/ revolutions of the pump handle. The flow rate shall be based upon the time to pump 1.5 gallons of oil. The pumping rate (cycles per minute) shall be recorded in the rest report. At the start, during, and after the test, all lines and hoses shall be full of oil and the relieving of pressure in the discharge line as specified in 3.7.2.1 shall be demonstrated. The test shall be repeated four times, except the last two tests shall be at 2 $^{+0.1}$ gpm and $^{+0.05}_{-0.0}$ 0.25 gpm, respectively. The performance specified in 3.6 shall be demonstrated. The entire unit shall function in a satisfactory manner.

4.6.4 Leakage test. The tank and pump unit with oil in the supply and overflow tanks shall be turned upside down before and after the unit has been built up to the operating pressure. There shall be no evidence of oil leakage.

4.6.5 Calibrated sight glass tests.

4.6.5.1 Overflow tank test. The overflow tank shall be filled with oil with measured quantities of 1, 3, 5, and 7 quarts. The sight glass shall show the amount added to the 1-, 3-, 5-, and 7 quart intervals to within ± .125 pint. These data shall be recorded in the test report.

* 4.6.5.2 Supply tank test. The supply tank shall be filled to the 3-gallon level as indicated on the sight glass. The tank and pump unit shall then discharge oil from the end of the fill nozzle into graduated containers. The sight glass shall show the amount of oil pumped from the supply tank at the 2-, 3-, 5-, 7-, 9-, and n-quart intervals to within + .125 pint. These data shall be recorded in the test report.

* 4.6.6 Mobility tests. The unit (see 6.3.4) shall be towed by a vehicle having a pintle height of 10 to 30 inches at an average speed of 5 mph (maximum speed of 7.50 mph) for a distance of 5 miles over a course consisting of right-angle obstructions .75 to 1 inch in height by 1.--to 2 inch in width alternately spaced at 5 and 6-foot intervals at sixty angles to the direction of travel. At the completion of this test, the unit shall be inspected and there shall be no evidence of structural damage or loosening of any part of the unit. The unit, with the supply tank full of oil, shall then be dropped from a height of 1 foot onto a concrete floor until each of the resting points has impacted before the others three times during the test. The unit shall be inspected after each impact, and there shall be no evidence of structural damage or loosening of any part. At the completion of these tests, the unit shall be able to dispense 3 gallons of oil in accordance with 4.6.3 at the 1 gpm at 50 cycles/minute rate.

4.6.6.1 Tilt test. The unit shall remain in a natural upright position when subjected to a downward force of 50 pounds applied within 15 degrees of vertical at the center of the pump handle grip with the supply tank empty. No external means shall be utilized to maintain the unit in an upright position.

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* 4.6.7 Environmental tests. Unless otherwise specified, the environmental tests shall be conducted in accordance with MIL-STD-810. Unless otherwise specified herein, at the end of each test, the tank and pump unit shall dispense 3 gallons of oil in accordance with the procedure specified in 4.6.3 at the 1 gpm at 50 cycles/rein rate. An engineering evaluation may be performed in lieu of testing for the tests specified in 4.6.7.5, 4.6.7.6, and 4.6.7.7. The engineering, evaluation shall consist of a written detailed analysis of the tank and pump unit and all its parts and components to satisfactorily accomplish the required environmental conditions.

* 4.6.7.1 High temperature test. The unit shall be exposed to not less than 160°F for not less than 48-hours. After that exposure, the temperature shall be reduced to not less than 125°F. After the oil temperature in the supply tank has stabilized, the unit shall be removed from the environmental chamber and immediately the unit shall be able to dispense 3 gallons of oil at the 1 gpm at 50 cycles/minute rate.

* 4.6.7.2 Low temperature test. The unit shall be exposed to not more than -80°F for not less than 48-hours. After that exposure, the temperature shall be raised to -40°F. After the oil temperature in the supply tank has stabilized, the unit shall be removed from the environmental chamber and immediately the unit shall be able to develop not less than 300 psig deadhead oil pressure with not more than 50 pounds exerted on the pump handle then immediately the unit shall be able to deliver not less than 50 gpm with not more than 50 cycles/minute of the pump handle and with not more than 50 pounds exerted on the pump handle.

* 4.6.7.3 Dust test. The sand and dust test shall be conducted in accordance with procedure I of method 510.2. Exposure shall be 90 minutes with unit's most vulnerable side facing the blowing dust. The unit shall be operated dry at its 1 gpm stroking rate. At conclusion of the 90 minute exposure, dust shall be removed from the unit's exterior then the interior of the units supply tank shall be examined, Any evidence of dust is a test failure.

* 4.6.7.4 Rain test. The rain test shall be conducted in accordance with procedure II of method 506. During the 15 minute exposure, the unit shall be operated dry at its 1 gpm stroking rate. At conclusion of the 15 minute exposure, water shall be removed from the unit’s exterior, the interior of the unit's supply tank shall be examined. Any evidence of water is a test failure.

* 4.6.7.5 Fungus resistance test. The fungus resistance test shall be conducted in accordance with method 508.3.

* 4.6.7.6 Humidity test. The humidity test shall be conducted in accordance with method 507.2.

* 4.6.7.7 Salt-fog test. The salt-fog test shall be conducted in accordance with method 509.2.

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4.6.8 Endurance test. The tank and pump unit shall be connected in series with a test rig consisting of a restrictor to create a 5 psi differential above the tank static pressure at 1 gpm, flow rate and the necessary hose and fittings to return the fluid to the supply tank. The use of auxiliary equipment other than the restrictor, hose, and fittings is subject to the approval of the responsible engineering activity. The unit shall be operated for 60,000 pump cycles at a flow rate of not less than 1 gpm. The pump shall dispense its maximum capacity during each pump cycle, The fluid temperature shall not exceed 130°F during the test. Test time of less than 15 minutes shall not be credited as part of the endurance test. The endurance test shall be conducted prior to the dust and rain tests.

4.6.9 Reliability demonstration and test. Satisfactory completion of all tests specified herein without failure demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.9.1 Reliability demonstration continuation. If at the completion of the first article tests any failures have been observed, the endurance test (4.6.8) shall be continued and shall be considered the reliability test. Measurements and observations shall be continued as time accumulates on the item. An accept decision shall be made when the following inequality holds

$$t + \geq 1/2\ (X^2 0.1,\ 2r + 2)\ \text{MTBF}$$

or the accumulated test operating time must be equal to or greater than 1/2 times the value of Chi squared distribution at 0.1 risk of error and 2r plus 2 degrees of freedom times the specified MTBF. The symbol r is the number of accumulated relevant failures. Note that tables of Chi squared should be used that yield approximately 4.6 at 0.1 risk (0.9 confidence of a correct decision) and no failures (2r plus 2 equals 2).

4.6.9.2 Truncation. If a decision cannot be reached by the method specified in 4.6.9.1 before the accumulated endurance test operating time is three times the specified endurance test time, the test shall be truncated or terminated at that time unless continued testing is authorized by the procuring activity, If at test termination the inequality still does not hold (the right side is greater), the equipment shall be considered to have failed the test.

4.6.10 Maintainability verification. Item compliance with the maintainability requirements shall be verified in accordance with MIL-STD-471 with the following provisions:

a. The failure rates of individual parts and assemblies used for frequency of maintenance predictions may be estimated or predicted from reliability test or other data available, vendor's data or commitments, comparison to experience with similar items, or other sources available to the contractor. It is not intended that special part MTBF tests be conducted for maintainability although any data on critical or potentially troublesome parts tested for reliability purposes should be used.

b. It may not be feasible to simulate or induce failures in some parts in the equipment. For such failures, an engineering analysis can determine what the failure symptoms would be. The test monitor may describe these symptoms to the maintenance personnel. The diagnosis and repair time can then be measured as the maintenance personnel go

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through the actions they would normally take in response to such symptoms and necessary repairs. The test monitor can inform the maintenance persons when they have correctly Identified the failed part.

c. The contractor shall list all the probable maintenance actions that may be needed in service. The number of actions to be simulated shall be 0.2 of all those listed and shall be selected randomly from all those listed unless otherwise specified.

* 4.7 Packaging inspection. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 or the documents specified therein.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The tank and pump unit shall be preserved in accordance with method I of MIL-P-116. Exterior surfaces of the tank and pump unit shall be flushed with corrosion preventive oil conforming to MIL-C-8188 and drained. The fill and overflow hose shall be secured to the positive-holding devices with tape conforming to PPP-T-60. Nozzle adapters shall be packaged and cushioned within the storage box.

* 5.1.2 Commercial. The tank and pump unit shall be preserved and packaged in accordance with the requirements of ASTM D-3951.

* 5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2).

* 5.2.1 Level A. Each tank and pump unit, preserved and packaged as specified in 5.1, shall be packed in a wood-cleated plywood shipping container conforming to PPP-B-601, overseas type, grade A, style optional. Insofar as practicable, containers shall be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

* 5.2.2 Level B. Each tank and pump unit, preserved and packaged as specified in 5.1, shall be packed in a wood-cleated plywood shipping container conforming to PPP-B-601 domestic type, grade B, style optional or fiberboard shipping container conforming to PPP-B-636, class weather resistant, style optional. Insofar as practicable, containers shall be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to the box specification.

* 5.2.3 Commercial. The tank and pump unit shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Physical protection. Blocking, bracing, cushioning, and testing shall be as specified in MIL-STD-1186.

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* 5.4 Marking. In addition to any special marking required by the contract or order, each unit pack and exterior shipping container shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

6. NOTES

* 6.1 Intended use. The PMU-29/E tank and pump unit is intended for dispensing oil to all aircraft engine and gear boxes requiring pressure-type oil servicing with synthetic base lubricating oil. The unit also dispenses corrosion preventive oil.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification,

b. Applicable levels of preservation, packaging, and packing (see section 5. and 5.2).

c. Special shipment markings (see 5.4).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is the probability that material will perform a specified function without failure under given conditions for a specified period. In general, the instantaneous failure rate is fairly constant throughout the normal operating life of equipment. Under this condition, the probability of no failure for any particular time period decreases exponentially as a function of the length of the period and of the MTBF. This leads to the following relationship for minimum reliability, R min.

$$R \text{ min.} = e^{-t/MTBF}$$

where t is the time period or mission time

and MTBF is the minimum value demonstrated by test of the lower confidence limit on the MTBF described below,

6.3,2 Mean time between failure. In general, MTBF is the average or arithmetic mean of the operating time between failures or the mean time between failure. As used in this specification, the term MTBF means the minimum MTBF acceptable at the specified confidence. Note that this value, calculated from the demonstration data, is less than the point estimate of the true MTBF which would be the sum of the test operation time divided by the sum of the relevant failures. MTBF is often used as an expression for MCBF (where t must be expressed in cycles to obtain mission reliability). MTBF is also used interchangeably with MTTF which applies to items that are expended at first failure

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and are not repairable. If an exponential distribution of failures is assumed, the Chi squared relationship given in 4.6.9.1 is appropriate for a lower confidence limit on the estimated true MTBF and therefore as criteria to determine if the MTBF demonstrated equals or exceeds the MTBF specified as minimum. If no relevant failures are observed, at 0.9 confidence this becomes:

$$\text{MTBF (rein)}_{\text{demonstrated}} = \frac{\text{Accumulated test operating time}}{2.3}$$

6.3.3 Mission time. The mission time is 54 pump cycles and is determined by the time required for servicing a two-engine aircraft.

6.3.4 Unit. The unit is defined as the 3-wheel dolly with supply and over-flow tanks, components, and towing handle.

6.3.5 Tank and pump assembly. The tank and pump assembly is defined as the supply tank, pump assembly, and related components.

6.4 Abbreviations. In order to save space and to avoid spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

gpm - gallons per minute

Hg - mercury

min - minimum

mm - millimeter

mph - miles per hour

MCBF - mean-cycles-between-failures

MTBF - mean–time-between-failures

MTTF - mean-time-to-fail

psi - pounds per square inch

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

* 6.6 Subject term (key word) listing.

Aircraft turbine engine oil services

Jet engine oil services

PMU-29/E tank and pump unit

Tank and pump unit, jet engine oil servicing

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6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-38266C	Tank and Pump Unit PMU-29/E, Jet Oil Servicing, 3 Gallon Capacity, Portable

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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